

Appendix B

Project+ Formulas and Equations Reference

Understanding project management formulas is critical to your success as a project manager. Appendix B is a helpful reference tool with all the equations that are discussed in the course, grouped by lesson and topic order.

Critical Path Method Formulas

Term	Name	Formula	Meaning	Covered in
PERT	Program Evaluation and Review Technique Estimate (PERT Estimate)	$\text{PERT Estimate} = (O + 4M + P)/6$	The most realistic project timing estimate, with adjustments for worst- and best-case scenarios	Lesson 9, Topic A
ES	Early Start (ES)	$\text{ES} = \text{maximum EF of all direct predecessors}$	The soonest an activity can start because of how long it will take to complete predecessor activities	Lesson 9, Topic A
EF	Early Finish (EF)	$\text{EF} = \text{ES} + t$ <i>where t = activity duration [i.e., time]</i>	The earliest an activity can finish because of how long it takes to complete it	Lesson 9, Topic A
LS	Late Start (LS)	$\text{LS} = \text{LF} - t$ <i>where t = activity duration</i>	The latest an activity can start and still allow the project to finish on time	Lesson 9, Topic A
LF	Late Finish (LF)	$\text{LF} = \text{minimum LS of all successors}$	The latest an activity can finish and still allow the project to finish on time	Lesson 9, Topic A
TF	Total Float (TF)	$\text{TF} = \text{LS} - \text{ES}$ $\text{TF} = \text{LF} - \text{EF}$	The amount of time an activity can wait to start without impacting the total project timeline	Lesson 9, Topic A
FF	Free Float (FF)	$\text{FF} = \text{minimum ES of all direct successors} - \text{ES} - t$ <i>where t = activity duration</i>	The amount of time an activity can delay without encroaching on the early start of any subsequent activities	Lesson 9, Topic A

Term	Name	Formula	Meaning	Covered in
CP	Critical Path (CP)	Critical Path = List of all activities where (TF = 0) If TF = 0, add that activity to the Critical Path	The sequence of events with zero TF that cannot afford any delays at all	Lesson 9, Topic A

Earned Value Management Formulas

Term	Name	Formula	Meaning	Covered in
PV	Planned Value (PV)	$PV = (\text{Planned \& Complete}) \times (BAC)$	What you should have spent at this point in the schedule.	Lesson 13, Topic A
BAC	Budget at Completion (BAC)	<i>No formula - this is the project cost baseline.</i>	How much the project should cost.	Lesson 13, Topic A
EV	Earned Value (EV)	$EV = (\text{Actual \% Complete}) \times (BAC)$	How much of the project schedule is actually completed.	Lesson 13, Topic A
AC	Actual Cost (AC)	<i>No formula - this is the project's total cost to date.</i>	How much the project has cost so far.	Lesson 13, Topic A
CV	Cost Variance (CV)	$CV = EV - AC$	How above or below budget the project is.	Lesson 13, Topic A
CPI	Cost Performance Index (CPI)	$CPI = EV / AC$	Whether the project is within budget.	Lesson 13, Topic A
SV	Schedule Variance (SV)	$SV = EV - PV$	How far ahead or behind schedule the project is.	Lesson 13, Topic A
SPI	Schedule Performance Index (SPI)	$SPI = EV / PV$	Whether the project is ahead of or behind schedule.	Lesson 13, Topic A
ETC	Estimate to Complete (ETC)	Bottom-Up ETC = sum of all remaining activities <i>Use when the estimates are wrong</i>	The cost needed to finish the remaining work.	Lesson 13, Topic A
ETC	Estimate to Complete (ETC)	Budgeted Rate ETC = BAC – EV <i>Use when the source of variance is eliminated</i>	The cost needed to finish the remaining work.	Lesson 13, Topic A

Term	Name	Formula	Meaning	Covered in
ETC	Estimate to Complete (ETC)	CPI $ETC = [(BAC - EV) / CPI]$ <i>where</i> $CPI = EV / AC$ <i>Use when you expect cost variance</i>	The cost needed to finish the remaining work.	Lesson 13, Topic A
ETC	Estimate to Complete (ETC)	CPI and SPI $ETC = [(BAC - EV) / (CPI \times SPI)]$ <i>where</i> $CPI = EV / AC$ <i>and</i> $SPI = EV / PV$ <i>Use when you expect schedule variance</i>	The cost needed to finish the remaining work.	Lesson 13, Topic A
EAC	Estimate at Completion (EAC)	$EAC = AC + ETC$	What the total project will probably cost.	Lesson 13, Topic A
VAC	Variance at Completion (VAC)	$VAC = BAC - EAC$	How much the total project will be above or below budget.	Lesson 13, Topic A

